

Gas Adsorption: Explanation of Equation Symbols

Symbol	Explanation	Aspen Adsorption Base Units
a	Specific particle surface	m^2/m^3
a_{HX}	Specific heat exchanger surface	$\text{m}^2(\text{HX area})/\text{m}^3(\text{Bed})$
a_p	Specific particle surface per unit volume bed	$\text{m}^2(\text{Particle area})/\text{m}^3(\text{Bed})$
A	Area	m^2
AA	Placeholder variable used for Dubinin-Astakov isotherm evaluation	
b	Kinetic Langmuir factor	$1/\text{bar}$
c_{bk}	Bulk gas phase concentration	kmol/m^3
c_k	Molar concentration of component k	kmol/m^3
c_{msk}	Macropore gas phase concentration	kmol/m^3
c_{sol}	Concentration of solid phase reactant	kmol/kg
c_{pai}	Specific heat capacity of adsorbed phase	$\text{MJ}/\text{kmol}/\text{K}$
$c_{p,cat}$	Specific heat capacity of catalyst	$\text{MJ}/\text{kg}/\text{K}$
c_{pg}	Specific gas phase heat capacity at constant pressure	$\text{MJ}/\text{kmol}/\text{K}$
c_{ps}	Specific heat capacity of adsorbent	$\text{MJ}/\text{kg}/\text{K}$
c_{pW}	Specific heat capacity of column wall	$\text{MJ}/\text{kg}/\text{K}$
c_{vg}	Specific gas phase heat capacity at constant volume	$\text{MJ}/\text{kmol}/\text{K}$
D_B	Bed diameter	m
D_{efc}	Effective micropore diffusion coefficient	m^2/s
D_{efP}	Effective macropore diffusion coefficient	m^2/s

Symbol	Explanation	Aspen Adsorption Base Units
D_{ek}	Effective adsorbed phase diffusivity of component k	m^2/s
D_{ki}	Knudsen diffusion coefficient of component i	m^2/s
D_{mk}	Mean molecular diffusion coefficient of component k	m^2/s
$E_{act,k}$	Activation energy for Arrhenius relationship	MJ/kmol
E_{ik}	Radial dispersion coefficient of component k	m^2/s
E_{zk}	Axial dispersion coefficient of component k	m^2/s
f	Function	-
f_{eq}	Equilibrium (isotherm) relationship	-
H_{amb}	Wall-ambient heat transfer coefficient	$\text{MW}/\text{m}^2/\text{K}$
H_B	Height of adsorbent layer	m
H_i	Rate of change of heat of adsorbed phase	$\text{MJ}/\text{m}^3/\text{s}$
H_R	Combined heats of homogenous and heterogeneous reactions	$\text{MJ}/\text{m}^3 \text{ (Bed)}/\text{s}$
H_{Rcat}	Heat of catalytic reaction	MJ/kmol
H_{Rgas}	Gas phase heat of reaction	MJ/kmol
H_{Ti}	Heat of adsorption contribution to solid phase energy balance	$\text{MJ}/\text{m}^3/\text{s}$
H_w	Gas-wall heat transfer coefficient	$\text{MJ}/\text{m}^2/\text{s}$
ΔH_i	Heat of adsorption of component i	MJ/kmol
HTC	Gas-solid heat transfer coefficient	$\text{MJ}/\text{m}^2/\text{s}$
IP	Isotherm parameter, units depend on isotherm	
j	Colburn j-factor for heat or mass transfer	-
$J_{ads,k}$	Mass transfer rate of component k owing to adsorption	$\text{kmol}/\text{m}^3 \text{ (Bed)}/\text{s}$
$J_{cat, reac,k}$	Mass transfer rate of component k owing to heterogeneous catalytic reactions	$\text{kmol}/\text{m}^3 \text{ (Bed)}/\text{s}$
$J_{gas, reac}$	Mass transfer rate of component k owing to homogenous, gas phase reactions	$\text{kmol}/\text{m}^3 \text{ (Void)}/\text{s}$
J_k	Mass transfer rate of component k to/from adsorbent	$\text{kmol}/\text{m}^3 \text{ (Bed)}/\text{s}$

Symbol	Explanation	Aspen Adsorption Base Units
k_{0k}	Pre-exponential factor for Arrhenius relationship	m/s
k_{0Pk}	Pre-exponential factor for pressure dependent Arrhenius relationship	m/s
k_{fk}	Film mass transfer coefficient of component k	m/s
k_g	Gas phase thermal conductivity	MW/m/K
k_{gr}	Effective radial gas phase thermal conductivity	MW/m/K
k_{gr}^{dyn}	Dynamic contribution to k_{gr}	MW/m/K
k_{gr}^{stat}	Static contribution to k_{gr}	MW/m/K
k_i	Effective, lumped mass transfer coefficient of component i	1/s
k_s	Solid thermal conductivity	MW/m/K
k_{gz}	Effective axial gas phase thermal conductivity	MW/m/K
k_{sr}	Effective radial solid phase thermal conductivity	MW/m/K
k_{sr}^{stat}	Static contribution to k_{gr}	MW/m/K
k_{sz}	Effective axial solid phase thermal conductivity	MW/m/K
k_w	Thermal conductivity of column wall	MW/m/K
K_{Ki}	Isotherm slope of component i (Henry's coefficient)	m ³ /kg
$\bar{K}_{Ki}$	Dimensionless isotherm slope of component i (Henry's coefficient)	-
K_{mac}	Macropore mass transfer coefficient	1/s
K_{mic}	Micropore mass transfer coefficient	1/s
K_p	Darcy's constant	bar s/m ²
K_{pi}	Macropore diffusion coefficient	m ² /s
L	Length of horizontal bed	m
M	Molecular weight	kg/kmol
MTC_g	Gas film mass transfer coefficient	1/s

Symbol	Explanation	Aspen Adsorption Base Units
MTC_s	Solid film mass transfer coefficient	1/s
ρ	Emissivity in calculation of effective thermal conductivities	
P	Pressure	bar
P_k	Partial pressure of component k	bar
p_i^0	IAS vapor pressure	bar
P_{sat}	Saturation pressure	bar
Q_{Hx}	Heat transfer rate to internal heat exchanger	MJ/m ² /s
r	Radial co-ordinate (in packed bed or particle)	m
r_c	Microparticle (crystal) radius	m
r_p	Particle radius	m
R	Universal gas constant	bar m ³ /kmol/K
R_{cat}	Catalytic reaction rate	kmol/kg/s
R_{gas}	Gas phase reaction rate	kmol/m ³ /s
R_{sol}	Solid phase reaction rate	kmol/kg/s
t	Time	s
t_{cycle}	Adsorption cycle time	s
T	Temperature	K
T_0	Equal to T_{amb} or T_W depending on context used	K
T_{amb}	Ambient temperature	K
T_c	Critical temperature	K
T_{CW}	Cooling water temperature	K
T_s	Solid phase temperature	K
T_g	Gas phase temperature	K
T_{Hx}	Heat exchange medium temperature	K

Symbol	Explanation	Aspen Adsorption Base Units
T_{St}	Steam temperature	K
T_W	Wall temperature	K
T_{ort}	Adsorbent tortuosity	-
U_{Hx}	Overall heat transfer coefficient: gas to heat exchange medium	MW/m ² /K
$U_{Hx,cw}$	Overall heat transfer coefficient: gas to cooling water	MW/m ² /K
$U_{Hx,St}$	Overall heat transfer coefficient: gas to steam	MW/m ² /K
v_g	Gas phase superficial velocity	m/s
w_k	Loading	kmol/kg
w_k^0	Pure component loading of component k	kmol/kg
W	Width of horizontal bed	m
W_T	Width of column wall	m
x_{char}	Characteristic length	m
x_k	Mole fraction of component k in the adsorbed phase	-
y_k	Mole fraction of component k in the gas phase	-
z	Axial co-ordinate	m
Z	Gas compressibility factor	-

Symbol	Explanation	Aspen Adsorption Base Units
a_{rg}	Radiation contribution to k_{gr}^{stat}	
a_{rs}	Radiation contribution to k_{sr}^{stat}	
β	Factor used in k_{gr}^{stat} calculation	
Δr	Radial discretization distance	m
ε_B	Total bed voidage	m ³ (Void+Pore)/m ³ (Bed)
ε_i	Interparticle voidage	m ³ (Void)/m ³ (Bed)

Symbol	Explanation	Aspen Adsorption Base Units
ε_p	Intraparticle voidage	m ³ (Pore)/m ³ (Particle)
ϕ	Function of packing density, used in k_{sr}^{stat} calculation	
ϕ	Relative pressure: $P_k / P_{sat,k}$	-
γ	Factor used in k_{sr}^{stat} calculation	
γ_i	Activity coefficient of component i	-
μ	Dynamic viscosity	N s/m ²
$\mu_{ads,i}$	Chemical potential of component i in the adsorbed phase	MJ/kmol
$\mu_{gas,i}$	Chemical potential of component i in the gas phase	MJ/kmol
ν_{jk}	Stoichiometric coefficient of component k in reaction j	-
Π_i^0	Spreading pressure of component i	bar m
θ	Time constant for adsorption cycle	-
ρ_g	Gas phase molar density	kmol/m ³
ρ_s	Adsorbent bulk density	kg/m ³
$\rho_{s,cat}$	Catalyst bulk density	kg/m ³
ρ_w	Wall density	kg/m ³
Ω	Parameter in Glueckauf expression	-
ψ	Particle shape factor	-

Dimensionless number	Defining expression	Description
Nu_W	$\frac{H_w x_{char}}{k_g}$	Nusselt number for gas wall heat transfer
Pe_H	$\frac{x_{char} v_g \rho_g M C_{pg}}{k_g}$	Gas-wall heat transfer Peclet number
Pe_K	$\frac{v_g H_b}{E_z}$	Component Peclet number for mass transfer

Dimensionless number	Defining expression	Description
Pr	$\frac{\mu C_{pg}}{k_g M}$	Prandtl number
Sc_k	$\frac{\mu}{D_{mi} \rho_g M}$	Component Schmidt number
Sh_k	$\frac{k_{fi} 2r_p}{D_{mi}}$	Component Sherwood number
Re	$\frac{2r_p M \rho_g v_g}{\mu}$	Particle Reynolds number